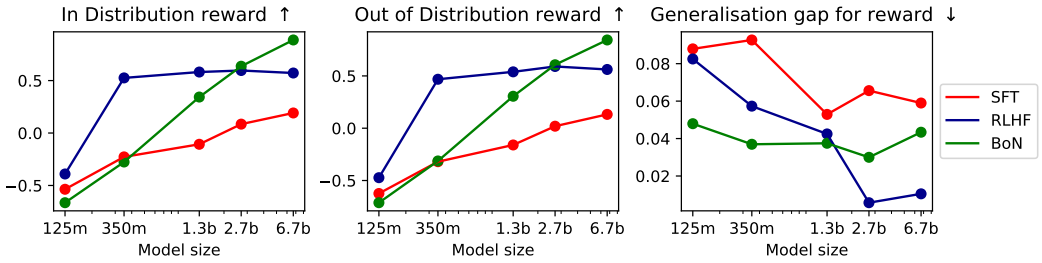

Figure 17: Proxy RM Score for SFT, BoN and RL models, averaged over dataset splits, for both in-distribution and out-of-distribution performance, and the generalisation gap. Arrows ↑, ↓ indicate whether higher or lower scores are better.

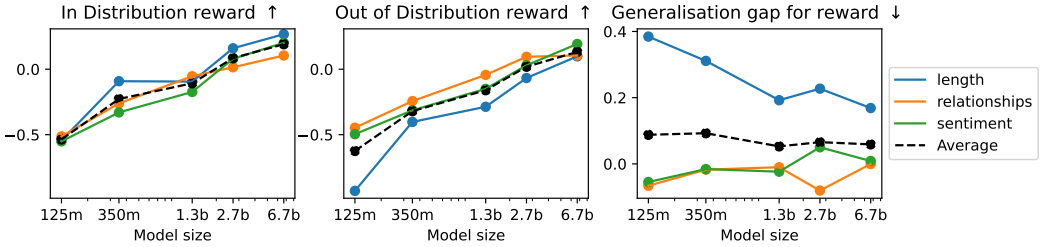

Figure 18: Proxy RM Score for SFT models for each dataset split, both in-distribution and out-of-distribution performance, and the generalisation gap. Arrows ↑, ↓ indicate whether higher or lower scores are better.

J.3 Generalisation Evaluation

For evaluation in these experiments, we train an RM as described in Appendix E.1 using the full dataset of summaries and preferences without splitting and the 6.7 billion parameter OPT model. We then use this *proxy RM* to evaluate the performance of SFT, BoN and RLHF models. As this reward model is trained with a different random seed and a different data distribution, it serves as a held-out automated evaluation of models, and a good proxy for human preferences.

We first discuss the results from the OPT version of the generalisation experiments described in Section 5.1. Fig. 17 shows the proxy RM score for SFT, BoN and RLHF models, averaged over dataset sizes. The important result here is that BoN generalises better than RLHF, which generalises better than SFT.

We see that at middling model sizes, RLHF outperforms BoN, but BoN scales better than RLHF, eventually outperforming it for models with more than 2.7b parameters. SFT comes out worst in this comparison, both scaling worse than BoN and the same as RLHF, and having worse absolute performance and generalisation. We see that BoN sampling does not see diminishing returns as model size increases, implying it will continue to be a useful yet simple technique. This experiment highlights the importance of training a reward model on human feedback and using it to select the best outputs at test time, potentially after fine-tuning the model.

Supervised Fine-Tuning. Fig. 18 shows the proxy RM score for the SFT models. Performance increases as model size increases, and in general performance drops OOD, which is unsurprising. There is a slight downward trend in the average generalisation gap across dataset splits as model size increases, implying that larger models fine-tuned with SFT generalise better (in terms of automated metrics). The relationships and sentiment splits produce negligible generalisation gap for the proxy RM score while the length split is more difficult.

Best-of-N. Fig. 19 shows the proxy reward score for the BoN models. Here we can see a smooth almost linear increase in performance as the model size increases. Given that the RM scores for smaller models were below chance, the fact the even for smaller models BoN results improve performance requires explanation. We hypothesise that the smooth increase here comes from two factors: the improvement in the SFT model being sampled from, and the improvement from the

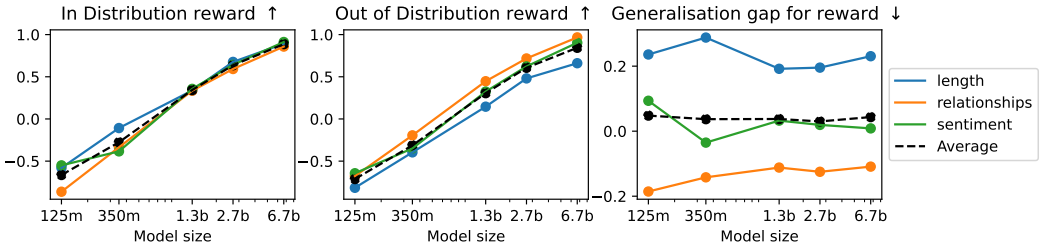

Figure 19: Proxy RM Score for BoN models for each dataset split, both in-distribution and out-of-distribution performance, and the generalisation gap. Arrows ↑, ↓ indicate whether higher or lower scores are better.

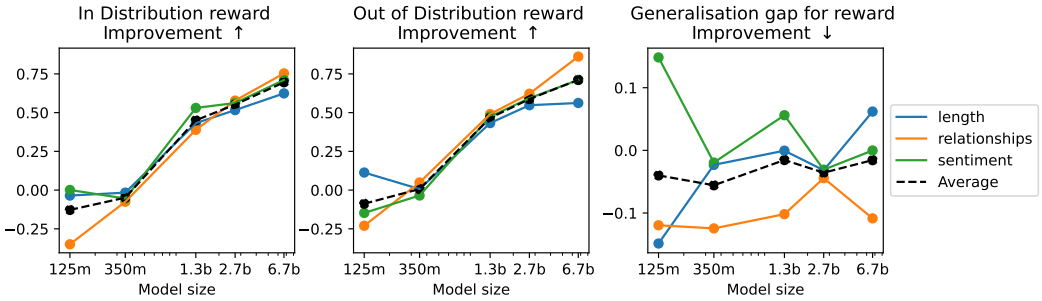

Figure 20: Proxy RM Score improvement using BoN on top of SFT models for each dataset split, both in-distribution and out-of-distribution performance, and the generalisation gap. Arrows ↑, ↓ indicate whether higher or lower scores are better. This plot is highlights the improvement from Fig. 18 to Fig. 19.

RM. At smaller model sizes increasing the number of parameters leads to improvements in SFT performance but not RM performance, while at larger model sizes increasing the number of parameters leads to improvements in both SFT and RM performance, but both to a lesser extent.

Fig. 20 shows the improvement of BoN over SFT. We can see that BoN only starts improving SFT as model size passes 350 million parameters, and the improvement grows as model size grows. This implies that as we increase model size BoN is likely to become a more performant choice compared to SFT. Further, we note that BoN uniformly improves the generalisation across model sizes (as shown by the black dashed line), implying that scaling models, even using SFT+BoN, will still result in a non-zero generalisation gap.

Reinforcement Learning from Human Feedback. Fig. 21 shows the proxy RM score for the RLHF models. Again, we see that increasing model size improves performance. Here we see a clearer trend of increasing model sizes reducing generalisation gap – this implies that as we make RL models larger, they are likely to generalise better. Given the difference between this trend and the generalisation gap trend for SFT models in Fig. 18, this partially justifies why RLHF is used in fine-tuning LLMs at a very large scale (Glaese et al., 2022; OpenAI, 2023; Ouyang et al., 2022): RLHF produces better-generalising models at larger model sizes than SFT.

J.4 Diversity Evaluation

Table 11 shows the per-input diversity scores (Eq. (2)) for both RLHF and SFT models. The RLHF models have much lower diversity than SFT models according to all three metrics. While RLHF leads to better-generalising policies, those policies generate much less diverse outputs. We also see that diversity does not seem to change much with model size, apart from a slight downward trend for diversity in SFT models as model size increases.

Table 12 shows the across-input diversity scores (Eq. (3)). Here the corpus of text over which the diversity is measured is a single input sampled from the model for a range of outputs. Even if a model produces less diverse outputs for a single input, it could still produce different inputs for different outputs. This is the case for the EAD score, which is a proxy for diverse vocabulary and syntax, as

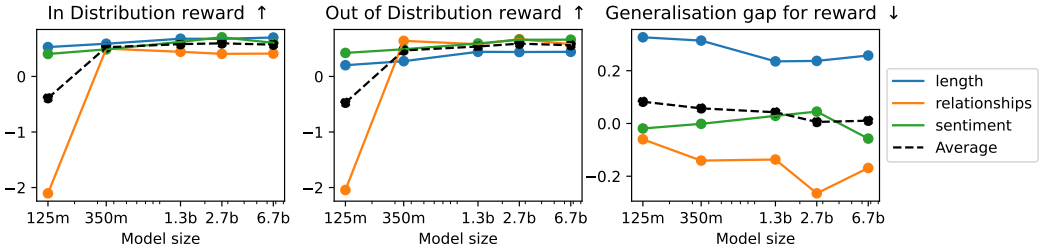

Figure 21: Proxy RM Score for RL models for each dataset split, both in-distribution and out-of-distribution performance, and the generalisation gap. Arrows ↑, ↓ indicate whether higher or lower scores are better.

Table 11: Per-input diversity scores for both RLHF and SFT models averaged over dataset splits. For these scores the outputs used to calculate the diversity are a sample of outputs from the model for single input. These per-input scores are then averaged, as in Eq. (2). Bolded results are better scores for each model size.

Model Size	125m		350m		1.3b		2.7b		6.7b	
Model Type	RLHF	SFT	RLHF	SFT	RLHF	SFT	RLHF	SFT	RLHF	SFT
EAD	0.15	0.81	0.15	0.8	0.13	0.81	0.16	0.8	0.07	0.79
Sent BERT	0.15	0.5	0.12	0.46	0.15	0.48	0.13	0.45	0.06	0.45
NLI	-1.08	0.26	-0.96	0.2	-1.0	0.12	-1.07	0.09	-1.54	0.06
Average	-0.26	**0.52**	-0.23	**0.49**	-0.24	**0.47**	-0.26	**0.45**	-0.47	**0.44**

Table 12: Across-input diversity scores for both RLHF and SFT models averaged over dataset splits. For these scores the outputs used to calculate the diversity are a set of single outputs from a range of inputs, as in Eq. (3). Bolded results are better scores for each model size.

Model Size	125m			350m			1.3b			2.7b			6.7b		
Model Type	RLHF	SFT	BoN	RLHF	SFT	BoN	RLHF	SFT	BoN	RLHF	SFT	BoN	RLHF	SFT	BoN
EAD	0.69	0.86	0.85	0.87	0.87	0.86	0.86	0.87	0.86	0.86	0.87	0.86	0.87	0.87	0.86
Sent BERT	0.63	0.73	0.72	0.71	0.73	0.71	0.68	0.73	0.71	0.7	0.74	0.71	0.73	0.73	0.71
NLI	0.05	0.35	0.32	0.2	0.38	0.3	0.25	0.36	0.27	0.3	0.28	0.27	0.32	0.32	0.3
Average	0.46	**0.64**	0.63	0.59	**0.66**	0.62	0.6	**0.65**	0.61	0.62	0.63	0.62	0.64	0.64	0.62

Table 13: Across-input diversity scores for BoN models averaged over dataset splits. For these scores the outputs used to calculate the diversity are a set of single outputs from a range of inputs, as in Eq. (3). This is equivalent to Table 2 in the main paper but for BoN policies. See response to reviewer hdr6 for more details.

Model Size	125m	350m	1.3b	2.7b	6.7b
EAD	0.85	0.86	0.86	0.86	0.86
Sent BERT	0.72	0.71	0.71	0.71	0.71
NLI	0.32	0.3	0.27	0.27	0.3
Average	0.63	0.62	0.61	0.62	0.62

Table 14: SFT Model ROUGE1 and reward model accuracy for the 3 smallest OPT model sizes, comparing freezing 80% of layers vs no layer freezing. Freezing generally results in less performance, as expected.

Model Size	SFT Model Rouge1		RM Accuracy	
	80% Frozen	0% Frozen	80% Frozen	0% Frozen
125m	0.217	0.221	0.482	0.496
350m	0.2241	0.2233	0.498	0.508
1.3b	0.221	0.2347	0.538	0.559

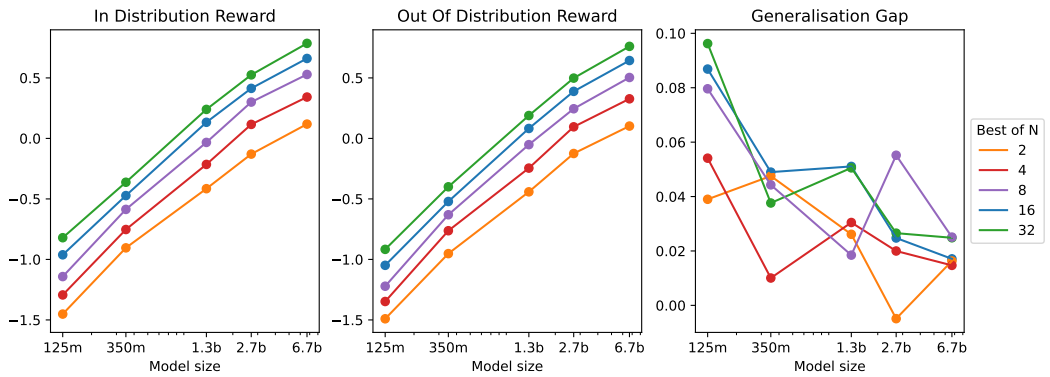

Figure 22: Proxy RM score for BoN sampling with varying N. All metrics are averaged over the 3 dataset splits.

for models with more than 125 million parameters both SFT and RLHF produce policies with very similar scores.

However, for the Sent BERT score, which is a proxy for diverse content and semantics, RLHF models are consistently less diverse than SFT models. This implies that RLHF produces models that have a tendency to generate outputs about certain topics or content regardless of the input. For the NLI score, which is a proxy for logical diversity, we see that as RLHF model size increases the score increases, eventually reaching the diversity of SFT models. Low NLI score for smaller models implies they have a tendency to make logically consistent claims in their outputs on top of producing outputs about certain topics of content.

J.5 Model Freezing Experiments

We perform a small experiment to evaluate the effects of freezing the first 80% of layers during fine-tuning. The results are shown in Table 14, and show that while performance drops for the models evaluated in the experiment, the drop is not catastrophic, justifying the use of model freezing.

J.5.1 BoN Performance for Different N

In the previous OPT results we use $N = 64$ samples in best of N sampling. Here we show proxy RM scores for $N = 2, 4, 8, 16, 32$, to show how choices of N trade off against performance. Fig. 22 shows proxy RM score in-distribution, out-of-distribution and the generalisation gap for these different choices of N. We see that increasing N does lead to improved performance. At lower model sizes it leads to a larger generalisation gap, but this does not hold for larger model sizes.

K RLHF and RM Training Curves

Here we present training curves for PPO and reward model training, for the summarisation task, for the models used in the main paper. In Fig. 23 we show the reward model validation accuracy throughout training. This is 1 epoch of training. In Fig. 24 we show the KL divergence and reward model score throughout training. PPO training has converged by approximately 250 PPO training steps, and so we terminated training early to save compute.